

# Samuel Monteiro Junior

## DEVOPS & PLATFORM ENGINEER

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### PROFESSIONAL PROFILE

Software Engineer with 4+ years of experience in infrastructure automation, Linux systems administration, and network security. Day to day, I provision servers with Docker and Docker Compose, configure secure mesh networks with Tailscale and WireGuard, and build APIs in Python (FastAPI) and Node.js. Focused on Site Reliability Engineering (SRE), operational hardening, and eliminating repetitive manual work through code and telemetry.

### TECHNICAL SKILLS

**DevOps & Infrastructure:** Docker, Docker Compose, Linux/Unix SysAdmin (Debian, Arch Linux, macOS), Caddy Server (Reverse Proxy, Automatic SSL), Mesh VPN (Tailscale, WireGuard), SSH Hardening, Networking & Firewalls (IPTables), Git

**Edge & Serverless:** Cloudflare Workers, Durable Objects, KV Store, D1, DNS-over-HTTPS, edge security heuristics

**Development & Automation:** Python (FastAPI, Web Scraping, CLI Scripts), Node.js (REST APIs, WebSockets, BullMQ, Redis), Rust (Tokio, Axum, Hyper), n8n Workflows, Shell/Bash Scripting for operational task automation

**Databases & Cryptography:** PostgreSQL, TimescaleDB, SQL Server, Redis Pub/Sub, SQLite, LUKS Full Disk Encryption, mTLS

### PROFESSIONAL EXPERIENCE

#### Automation Engineer & Full Stack Developer · Freelancer

April 2022 to Present

- Architected and deployed microservices ecosystems using FastAPI and Node.js in Docker containers, reducing request processing times by 40%.
- Designed and implemented a private mesh network (Tailscale VPN) with mTLS authentication to connect local hosts to Debian VPS servers, securing sensitive data traffic.
- Developed Cortex-Nexus, an automation pipeline (n8n, Python, AI) for lead extraction and qualification, achieving 99.7% deliverability and processing 3,000+ leads per month.
- Structured Parana-Leads, with automated deployment in Docker containers using Caddy as a reverse proxy with automatic SSL, reducing deployment time for new environments by 70%.

#### Support Analyst / Database Administrator · CNI Tecnologia e Software

June 2024 to December 2024

- Administered Linux/Windows network security policies and servers, reducing security incidents by 25% through connection hardening and access control.
- Developed Bash and Python scripts to automate relational database backups and hardware performance telemetry, reducing manual monitoring effort by 15 hours per week.
- Optimized queries and maintenance routines in SQL Server and PostgreSQL databases, reducing management report load times by 50%.

#### Support Analyst & Web Developer · Vip Informática

February 2017 to January 2021

- Administered operational databases (PostgreSQL/SQL Server), implementing automatic backup routines and reducing downtime to near zero.
- Developed and maintained internal tools (Node.js/PHP) focused on query and load time optimization, accelerating internal API response times by 35%.
- Led the restructuring of IT support processes and technical documentation, resulting in a 30% reduction in repetitive support tickets.

## PROJECTS

### Aegis: Reverse Proxy and Rate Limiter

[Rust](#) / [Tokio](#) / [Axum](#) / [Docker](#)

Reverse proxy in Rust with per-IP token-bucket rate limiting, timeouts at every stage (including slowloris protection), SIGHUP configuration reload without dropping connections, Prometheus metrics on a separate admin plane, and a FROM scratch Docker image running as a non-root user.

### Sentinel: Edge Bot Firewall

[Cloudflare Workers](#) / [Durable Objects](#) / [JavaScript](#)

Edge bot firewall running on Cloudflare Workers and Durable Objects: each request is scored by TLS, HTTP version, and browser header signatures, a per-client adaptive rate limit, and a verdict right at the edge (allow, challenge, or block), with logging off the response path and within the free tier.

### Cortex-Vault: Knowledge Infrastructure

[Linux](#) / [Shell Script](#) / [Python](#) / [Tailscale](#)

High-performance data ingestion system, automatically synchronized via Tailscale VPN mesh between Arch Linux and macOS. Developed Shell and Python utilities for fast local search and indexing.

### BoxOptimizer API

[.NET 8](#) / [JWT](#) / [Docker](#)

High-performance API for three-dimensional logistics calculation (efficient box cubing) using the First Fit Decreasing (FFD) algorithm, packaged as a Docker image and ready for horizontal scaling.

## EDUCATION

### Associate's Degree in Systems Analysis and Development

2024, ongoing

PUCPR (Distance Learning)

### Bachelor of Laws (LL.B.)

2016 to 2021

UNISEPE

### CC50: Introduction to Computer Science

2023

Harvard University / Fundação Estudar